

VU THI HAI YEN

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ABOUT ME

Machine Learning Engineer | Applied Scientist | Data Scientist

Machine learning practitioner (current PhD Candidate) with 4+ years across industry and research, taking models from raw, messy data through to deployed services. My work spans graph and neuro-symbolic learning, learning from partially-labelled data, NLP, and mineral prospectivity mapping. Two blind peer-reviewed publications (ACM SAC 2025, ECML PKDD 2026), each beating state-of-the-art baselines. Strong Python/PyTorch; used to collaborating with non-ML domain experts.

WORK EXPERIENCE

Université d'Orléans / BRGM

Orléans, France

PhD Researcher, Machine Learning

October 2023 – Present

- Designed *NeuralAdjust*, a neuro-symbolic PU-learning method (NeuPSL, PyTorch) that dynamically reweights logical rules to resolve conflicting constraints; outperformed the state-of-the-art graph-based PU method (GPL) across benchmarks - e.g. F1 **88.1 vs 81.9** on Cora and **73.6 vs 36.2** on Chameleon, while improving class-prior estimation.
Accepted at ECML PKDD 2026.
- Developed a feature-mining graph-construction method for *mineral prospectivity mapping* on an imbalanced dataset (105 positives among 14,218 nodes, 118 features); *increased* recall@top-5% by **12.5%** (GCN) and **9%** (GAT) over the spatial-only baseline, beating XGBoost and MLP.
Published at ACM SAC 2025.
- Translated geoscientists' domain expertise into model constraints and features; delivered two blind peer-reviewed papers and presented results to non-ML stakeholders across 6+ talks and seminars.

ECMWF Code4Earth

Remote

ML Engineer - 5-person team

2024 – 2025

- Collaborated in a **team of 5**, mentored by ECMWF scientists, to deliver two open-science ML challenges from prototype to working application - coordinating across data pipeline, modelling, and deployment via shared Git workflows.
- DSS ML-Costing** (2025): co-built and deployed a cost-estimation service for ECMWF's Data Store - a FastAPI predicting request processing time. Trained XGBoost/LightGBM with Optuna + 5-fold CV across 18 datasets (up to **866K** samples), reaching $R^2 \approx 0.96$ and MAPE $\approx 13\%$ on the strongest datasets; contributed schema validation, feature engineering, and model evaluation.
- KGB-TruthGuiding** (2024): co-built a GraphRAG system to ground and visualize LLM answers over ECMWF scientific literature - LLM-based entity/relation extraction (GPT, Llama 3) into a knowledge graph, community detection with local/global retrievers; the team assembled a ~ 490 -document corpus via a custom crawler and PyMuPDF.

LIGM Lab, Gustave Eiffel University

Paris, France

Research Engineer / Research Intern

November 2021 – July 2022

- Shipped production features into Unitex/GramLab, an open-source NLP platform, implementing word-tag selection persistence across the C++ core engine and Java visual IDE.
- Built an English word-lattice dataset and a PyTorch model for lattice-structured NLP data.

SYSTRAN

Paris, France

NLP Research Intern

June 2020 – November 2020

- Developed contextualized paraphrase generation with tag-specific Transformer/GPT2 models
- Used the synthetic data to augment OpenNMT translation across English–French, English–German, and English–Vietnamese.

EDUCATION

Université d'Orléans - PhD Candidate in Machine Learning, in collaboration with BRGM	2023 – Present
École Polytechnique - Master 2 in Data Science; graduated with High Honors	2019 – 2021
Université Paris-Est Créteil - Master 1 in Mathematics and Applications	2018 – 2019
Hanoi National University of Education - Bachelor of Mathematics; Very Good Degree	2014 – 2018

PUBLICATIONS

Positive-Unlabeled Learning with Knowledge-Structured Data [code] Thi Hai Yen VU, Vincent NGUYEN, Christel VRAIN, Hugo BREUILLARD, and Thi Bich Hanh DAO. <i>European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases - ECML PKDD 2026 (acceptance rate 24%)</i>
Features Leverage in Graph Models for Mineral Prospectivity Mapping [link] Thi Hai Yen Vu, Thi Bich Hanh Dao, Vincent Nguyen, Christel Vrain, and Hugo Breuillard. <i>Proceedings of the 40th ACM/SIGAPP Symposium on Applied Computing, pp. 597–604 (acceptance rate 25%)</i>

ACADEMIC ACTIVITIES

Teaching – Université d'Orléans, bachelor level: Programming and Algorithms (16h practicals); Discrete Mathematics (18h tutorials + 12h practicals).
Talks & Presentations – ANR IA.iO Day; BRGM “Digital Data” seminar (poster); ARETHUSE webinar; BRGM Young Researchers AI Group; MaDiTraCe Workshop; CA team seminar. 2024 – 2025
Scientific Training – Nordic Probabilistic AI School; Eastern European Machine Learning Summer School; MaDICS Symposium; CNRS Fête de la Science (2024).

LANGUAGES & SKILLS

Programming	Python, Java, C++, R, MATLAB
ML / Deep Learning	PyTorch, PyTorch Geometric, TensorFlow/Keras, scikit-learn, XGBoost, LightGBM
ML Methods	PU & semi/self-supervised learning, graph learning (GCN/GAT), neuro-symbolic (NeuPSL/PSL), clustering, NLP, LLMs & GraphRAG
Data & Libraries	pandas, NumPy, NetworkX, geopandas, PyMuPDF, OpenNMT
Natural Languages	Vietnamese (native), English (proficient), French (upper-intermediate / B2)

ACADEMIC HONORS

Eiffel Excellence PhD Scholarship – French Ministry for Europe and Foreign Affairs	2022
Masters Scholarship – Foundation of France	2018 – 2020
Dam Phu My Scholarship for Excellent Students; NKTP Global Award	2016
NKTP Global Award	2014